

LIGHTWEIGHT REMOTE SENSING SEGMENTATION UTILISING JPEG AI LATENTS

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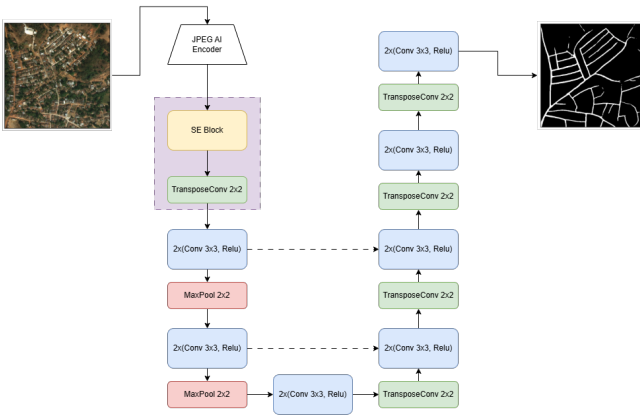


Fig. 1. JPEG AI Latent UNet Architecture

Earth Observation missions generate satellite imagery at scales that outpace conventional pixel-domain processing pipelines; enforcing full reconstruction before inference adds unnecessary overhead, despite image data being routinely compressed for transmission. JPEG AI [1] produces latent representations that encode features that are applicable to vision tasks. While latent domain inference has been demonstrated for face detection [2], semantic segmentation of remote sensing (RS) imagery, a more demanding task, remains unexplored. We show that a network operating directly on JPEG AI latents can outperform a baseline UNet while utilising fewer parameters.

The JPEG AI analysis transform encodes hierarchical spatial structure analogous to the upper encoder stages of a UNet [3]; we remove those stages and operate directly on the latent (Fig. 1). A squeeze-and-excitation block at the bridge reweights feature channels by task relevance, yielding the architecture shown. The model trains on latents from all four JPEG AI quality levels of the Massachusetts Roads dataset [4].

The JPEG-AI Latent-UNet achieves an IoU of 0.5439 and F1 of 0.6994, outperforming the pixel-domain baseline across

Table 1. UNet baseline vs. JPEG AI Latent UNet

Model	Params	GFLOPs	IoU	F1	ms/tile
Baseline	31.0 M	267.59	0.491 ± 0.122	0.650 ± 0.138	43.05
Latent	3.1 M	51.74	0.543 ± 0.094	0.699 ± 0.085	9.63

all metrics (Table 1). The considerably tighter metric spread relative to the baseline suggests more consistent road delineation across scene types, implying JPEG-AI latents encode more stable features than reconstructed imagery. The model also generalises across bitrates: a BPP sweep across all four quality levels produces a reliable accuracy-rate curve, demonstrating effective operation across varied operational bitrates.

These results validate compressed-domain inference as an effective direction for RS segmentation, delivering improved accuracy at an efficient cost. Beyond ground-based pipelines, this approach has longer-term implications for cases such as onboard processing.

1. REFERENCES

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